



# Append HMI Studio

## User Manual

Designing, testing and deploying HMI applications,  
illustrated with the LiquidWeighHMI example

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Append Automation

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# 1 Introduction

Append HMI Studio designs operator screens (HMIs) for PLCs, runs them against live equipment and publishes them for the plant floor. This manual shows how to build an HMI application with it, step by step, using a complete, working example: **LiquidWeighHMI**, the operator interface of a liquid weigh-up system that meters three plasticizers into a weigh hopper and discharges each batch to a mixer.

The example is included with Append HMI Studio as `examples/LiquidWeighHMI.ahmi`. Open it with **File > Open** and follow along; every screenshot in this manual was taken from it, the run-mode ones with the application connected to its ControlLogix controller.

*Table 1. What the LiquidWeighHMI example contains*

| Part          | In the example                                                                     |
|---------------|------------------------------------------------------------------------------------|
| Target screen | 1920 × 1080, full screen                                                           |
| PLC           | One ControlLogix controller (EtherNet/IP), 419 I/O tags                            |
| Screens       | Process, Batch, Supply, Devices, Setpoints, Alarm History and IO Sim               |
| Popup windows | A faceplate for each of 11 valves and 3 pumps, and 4 confirmation dialogs          |
| Alarms        | 34 alarmed tags, an Active Alarms list on every screen and an Alarm History screen |
| Security      | Three users (Operator, Maintenance, Engineer) with access levels                   |
| Scripts       | Window scripts that turn the sequence step number into text                        |

## 1.1 How an HMI application is organized

An Append HMI Studio project (.ahmi file) holds everything the operator station needs:

- **Devices:** the PLCs the application talks to (Allen-Bradley Logix, SLC 500 and MicroLogix, Modbus TCP, or the built-in simulator).
- **Tags:** named values, either read from and written to a PLC (I/O tags) or kept by the HMI itself (memory tags). Tags carry engineering units, ranges and alarm limits.
- **Windows:** each page of the drawing is a window of the application, either a full screen or a popup such as a faceplate.
- **Objects and animation links:** the shapes on the screens, and the links that tie them to tags: colors that follow a state, levels that fill, values that display, buttons that write.
- **Application settings:** the target screen resolution, the windows that open at start-up, how the published application runs, and its users.

The usual order of work, which this manual follows, is: application settings, devices, tags, screens and windows, animation, alarms and security, then testing with Run and deploying.

## 2 The Studio window

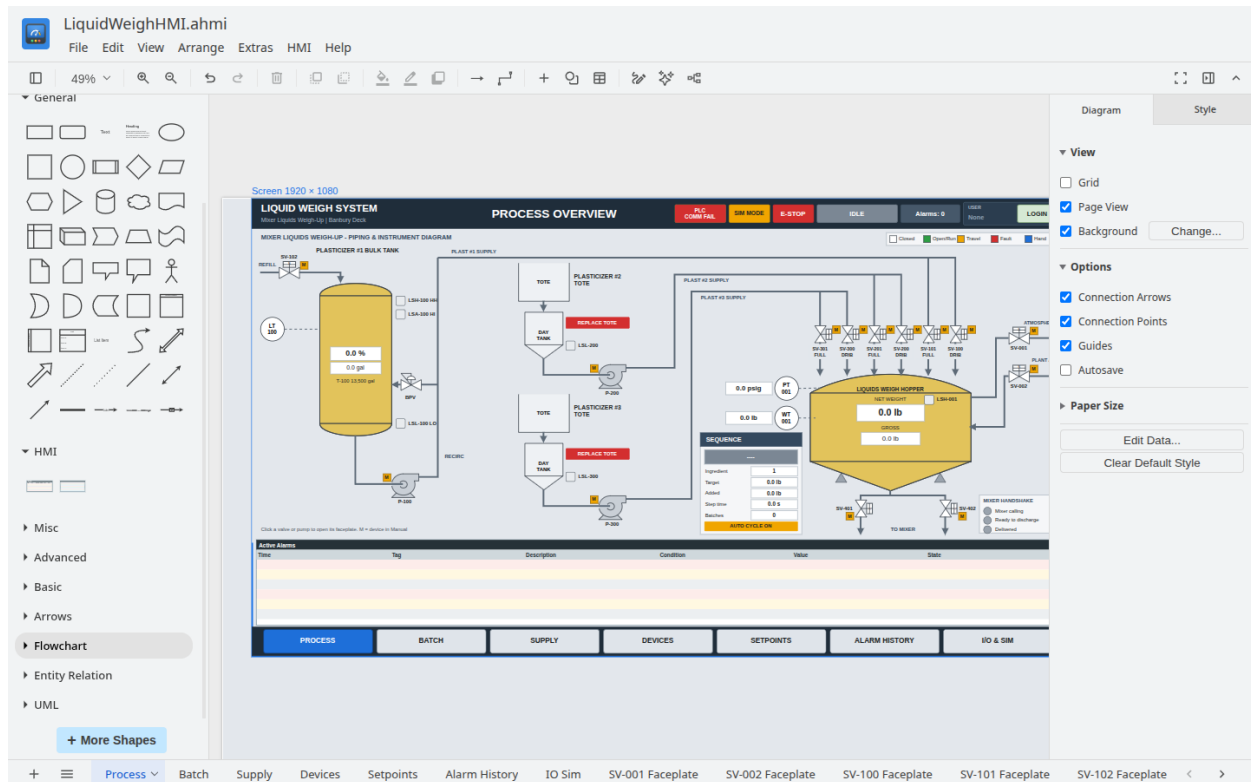


Figure 1. Append HMI Studio with the LiquidWeighHMI Process screen open

Table 2. Parts of the Studio window

| Area                | What it is for                                                                                                                                                       |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Menu bar</b>     | File, Edit, View, Arrange and Extras from the drawing editor, plus the HMI menu for everything specific to HMI applications                                          |
| <b>Toolbar</b>      | Zoom, undo, delete, arrange, and quick access to fill, line and shadow                                                                                               |
| <b>Shapes</b>       | Shape libraries (General, Misc, Advanced and more under More Shapes, including the P&ID libraries) and the HMI palette with the Alarm List and Alarm History objects |
| <b>Canvas</b>       | The page being edited. The dashed Screen frame shows the target screen (1920 × 1080 here)                                                                            |
| <b>Format panel</b> | Style, Text and Arrange for the selected object, and the Animation tab for its links                                                                                 |
| <b>Page tabs</b>    | One tab per window of the application; + adds one                                                                                                                    |

## 2.1 The HMI menu

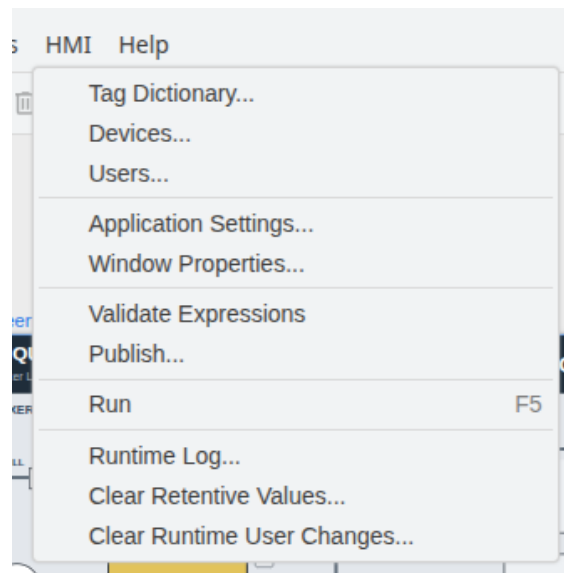


Figure 2. The HMI menu

Table 3. HMI menu commands

| Command                       | Use                                                                           |
|-------------------------------|-------------------------------------------------------------------------------|
| Tag Dictionary...             | Create and edit tags (chapter 5)                                              |
| Devices...                    | Configure the PLCs (chapter 4)                                                |
| Users...                      | Users, passwords and access levels (chapter 10)                               |
| Application Settings...       | Target screen, start-up windows, runtime and security settings (chapter 3)    |
| Window Properties...          | Type, position, size and scripts of the current window (chapter 6)            |
| Validate Expressions          | Check every link, script and address for errors (chapter 11)                  |
| Publish...                    | Build a Windows installer of the application (chapter 12)                     |
| Run / Stop (F5)               | Run the application in the Studio against live or simulated data (chapter 11) |
| Runtime Log...                | Messages from the last runs: connections, writes, script errors               |
| Clear Retentive Values...     | Forget the saved values of retentive tags on this computer                    |
| Clear Runtime User Changes... | Forget user changes made while running on this computer                       |

## 3 Application settings

Start a new application by setting its target screen: the resolution of the operator station it will run on. Open **HMI > Application Settings**.

**Application Settings**

TARGET SCREEN

Resolution: 1920 × 1080

Width (px): 1920

Height (px): 1080

STARTUP WINDOWS

Opened in page order when the application starts, each on top of the last. A replace window closes any earlier one it overlaps. With none ticked, the page being edited opens.

|                                             |                                         |
|---------------------------------------------|-----------------------------------------|
| <input checked="" type="checkbox"/> Process | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> Batch              | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> Supply             | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> Devices            | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> Setpoints          | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> Alarm History      | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> IO Sim             | Replace, 1920 × 1080 at 0, 0            |
| <input type="checkbox"/> SV-001 Faceplate   | Popup, title bar, 460 × 560 at 730, 150 |

RUNTIME

How the published application runs on the target PC. Kiosk fills the screen and hides the taskbar; Windows keys and Ctrl+Alt+Del still work unless Windows itself is locked down. The exit shortcut is Ctrl+Alt+Shift+Q.

Cancel OK

*Figure 3. Application Settings: target screen and start-up windows*

- **Resolution:** choose a common size or type the width and height. LiquidWeighHMI targets a 1920 × 1080 panel PC. The dashed Screen frame on every page follows this size.
- **Startup windows:** the windows that open when the application starts, in page order. LiquidWeighHMI opens the Process screen. With none ticked, the page being edited opens.

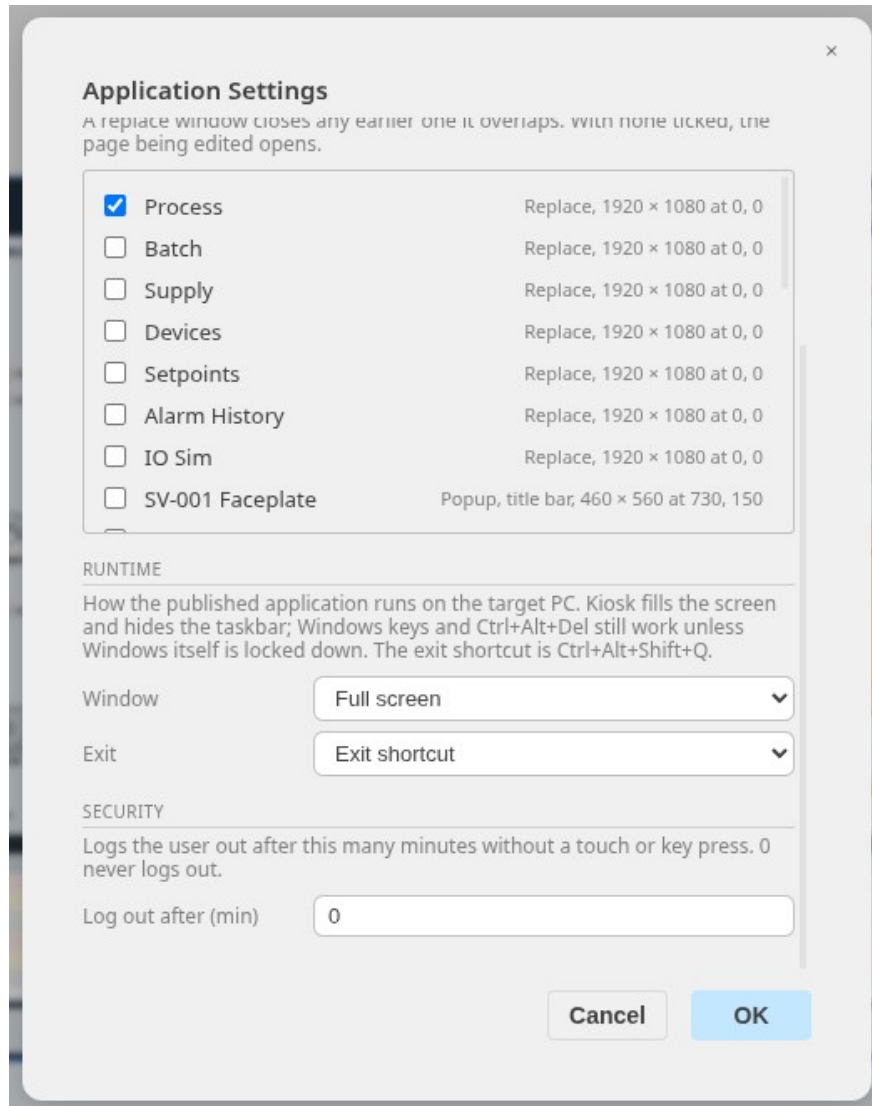


Figure 4. Application Settings: runtime and security

- **Window:** how a published application runs on the target PC: **Kiosk** (full screen, locked, taskbar hidden), **Full screen**, or a **window** at the target resolution. LiquidWeighHMI uses Full screen.
- **Exit:** the exit shortcut Ctrl+Alt+Shift+Q; the shortcut followed by a password; or never (the PC must be shut down to stop it).
- **Log out after (min):** logs the operator out after this many minutes without a touch or key press. 0 never logs out.

## 4 Devices

Devices are the PLCs the application communicates with. Open **HMI > Devices**, click **Add**, and fill in the connection. LiquidWeighHMI has one device, **LiquidWeighPLC**, a ControlLogix controller.

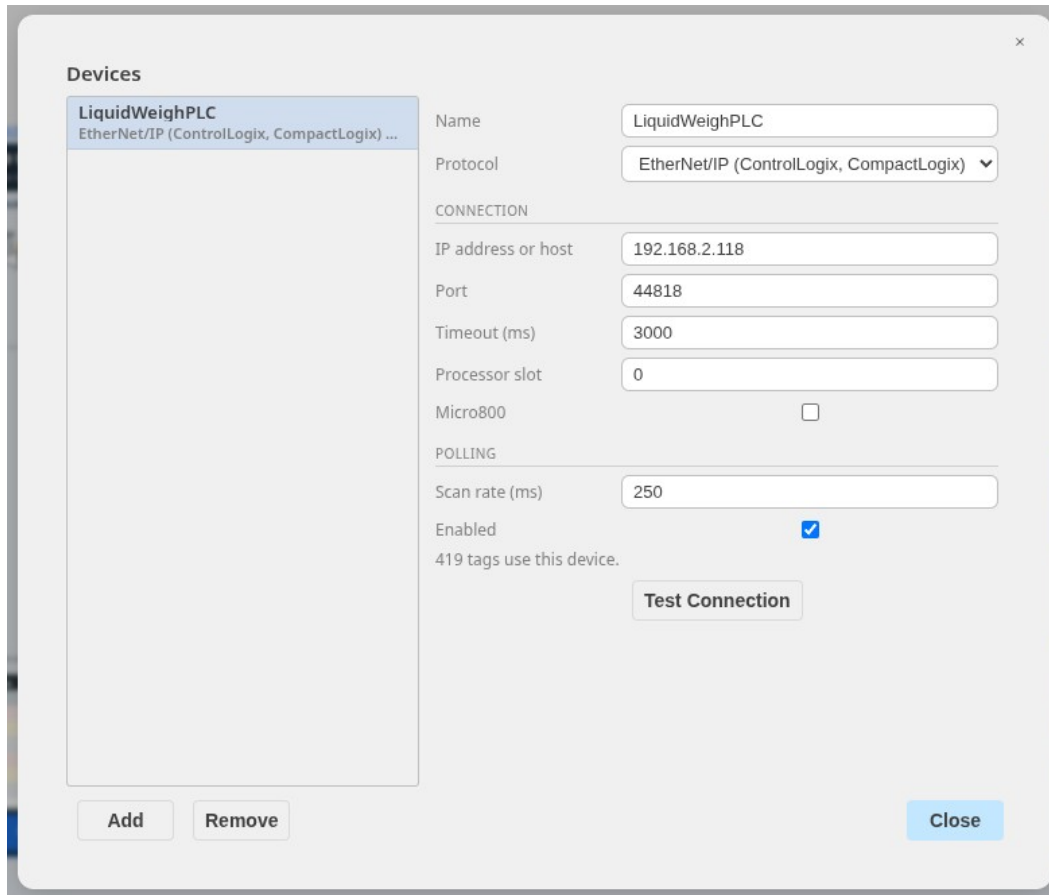


Figure 5. The Devices dialog with the LiquidWeighPLC controller

- **Protocol:** the communication driver (Table 4).
- **IP address or host, Port** (44818 for EtherNet/IP, 502 for Modbus TCP) and **Timeout**.
- **Processor slot:** the controller's slot in the chassis (0 for a CompactLogix).
- **Scan rate:** how often subscribed tags are read, 250 ms here.
- **Test Connection** checks that the PLC answers before any screen is built.

Table 4. Protocols and address examples

| Protocol                   | Controllers                          | Address examples                                               |
|----------------------------|--------------------------------------|----------------------------------------------------------------|
| <b>EtherNet/IP (Logix)</b> | ControlLogix, CompactLogix, Micro800 | PT_001.Value<br>Program:Main.Pump.Run<br>Recipe[3]<br>Status.5 |
| <b>PCCC (SLC)</b>          | SLC 5/05, MicroLogix                 | N7:0<br>B3:1/4<br>F8:2<br>T4:0.ACC                             |
| <b>Modbus TCP</b>          | Any Modbus TCP device                | HR:0<br>HR:10:FLOAT<br>CO:5<br>IR:3                            |

| Protocol  | Controllers     | Address examples             |
|-----------|-----------------|------------------------------|
| Simulator | None (built in) | Simulated values for testing |

**Note:** Communication runs through a small companion program, the hmi-comms server, which the Studio starts by itself. Device states and errors appear in HMI > Runtime Log.

## 5 Tags

Tags are the named values the screens show and change. Open **HMI > Tag Dictionary**. The list on the left shows every tag with its type; type in **Filter** to find one. Select a tag to edit it, or choose a type under **New tag...** to create one.

Figure 6. The Tag Dictionary with PT001, the weigh hopper pressure

Table 5. Tag types

| Type       | Holds                   | LiquidWeighHMI example                             |
|------------|-------------------------|----------------------------------------------------|
| IODiscrete | On/off from a PLC       | SV001_Opened (valve open limit)                    |
| IOInteger  | Whole number from a PLC | SV001_Status (0 closed, 1 open, 2 travel, 3 fault) |
| IOReal     | Real number from a PLC  | PT001 (hopper pressure, psig)                      |
| IOMessage  | Text from a PLC         | Recipe or product names                            |

| Type                                                     | Holds                   | LiquidWeighHMI example               |
|----------------------------------------------------------|-------------------------|--------------------------------------|
| MemoryDiscrete, MemoryInteger, MemoryReal, MemoryMessage | A value kept by the HMI | StepText (sequence step description) |

## 5.1 Tag fields

Table 6. Tag fields

| Field                                     | Meaning                                                                                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                      | Letters, digits and underscores; not case-sensitive. LiquidWeighHMI names tags after the device and signal, such as SV001_Status and PT001 |
| Comment                                   | A description. For an alarmed tag it is the alarm text operators see                                                                       |
| Engineering units, Minimum EU, Maximum EU | Units and range, used by fills, bars and value displays                                                                                    |
| Initial value                             | Starting value of a memory tag                                                                                                             |
| Device, Address                           | For an I/O tag: the PLC and the address in its syntax. The address is checked as you type (green when valid)                               |
| Scale raw values                          | Maps a raw integer range (for example 0 to 4095) to the engineering range                                                                  |
| Retentive                                 | For a memory tag: keeps its last value between runs                                                                                        |
| Simulation                                | A simulated value (sine, ramp, random, toggle) for testing without the PLC                                                                 |

## 5.2 Alarms on a tag

Scroll down to **Alarms** to give a tag alarm limits. PT001 alarms at High 70 psig and HiHi 85 psig, with a deadband of 2 so the alarm clears only below 68 psig. A discrete tag instead has **Alarm when** On or Off: SV001\_Fault alarms when it is on.

Figure 7. Alarm limits and deadband for PT001

Tags can also be exported to and imported from CSV (**Export CSV**, **Import CSV...**), which is the fastest way to create hundreds of tags from a PLC tag list: LiquidWeighHMI's 420 tags were entered that way.

## 6 Screens and windows

Each page of the drawing is a window of the application, named by its page tab. LiquidWeighHMI has seven full screens and eighteen popups.

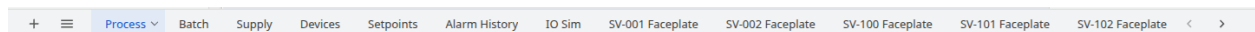


Figure 8. Page tabs: every page is a window

Table 7. Window types

| Type                 | Behavior                                                        | Used in LiquidWeighHMI for   |
|----------------------|-----------------------------------------------------------------|------------------------------|
| <b>Replace</b>       | Closes any window it overlaps when it opens                     | The seven main screens       |
| <b>Overlay</b>       | Opens on top and leaves other windows open                      | Information panels           |
| <b>Popup (modal)</b> | Stays on top; nothing beneath it can be touched until it closes | Faceplates and confirmations |

## 6.1 Window properties

Select a page and open **HMI > Window Properties**. For the SV-001 faceplate the window is a popup with a title bar, 460 × 560 pixels at 730, 150 on the screen. The preview shows where it appears.

The screenshot shows the 'Window Properties' dialog box for the 'SV-001 Faceplate' window. The dialog is titled 'Window Properties' and has a close button (X) in the top right corner. It is divided into several sections:

- Window:** A dropdown menu showing 'SV-001 Faceplate'.
- APPEARANCE:** A section with a 'Title bar' checkbox that is checked.
- TYPE:** A section with three radio button options: 'Replace' (unchecked), 'Overlay' (unchecked), and 'Popup (modal)' (checked). Below each option is a brief description of its behavior.
- POSITION AND SIZE (SCREEN IS 1920 × 1080):** A section with a descriptive text: 'The window shows the part of the page inside it – the dashed border on the canvas. Moving the window does not move what is drawn.' Below this are four input fields for 'Left (px)' (730), 'Top (px)' (150), 'Width (px)' (460), and 'Height (px)' (560). At the bottom of this section are three buttons: 'Full Screen', 'Fit to Content', and 'Center'.
- SCRIPTS:** A section with two text input fields. The first is labeled 'On show' and the second is labeled 'While showing'. Both fields contain the text 'QuickScript statements'.
- Buttons:** At the bottom right of the dialog are 'Cancel' and 'OK' buttons.

Below the input fields for position and size, there is a preview area showing a black rectangle with a smaller white rectangle inside it, representing the window's position and size on the screen.

Figure 9. Window Properties of the SV-001 Faceplate popup

A window shows the part of its page under its rectangle, so a popup's objects are drawn at the popup's position on its page. Figure 10 shows the SV-001 faceplate page in the editor.

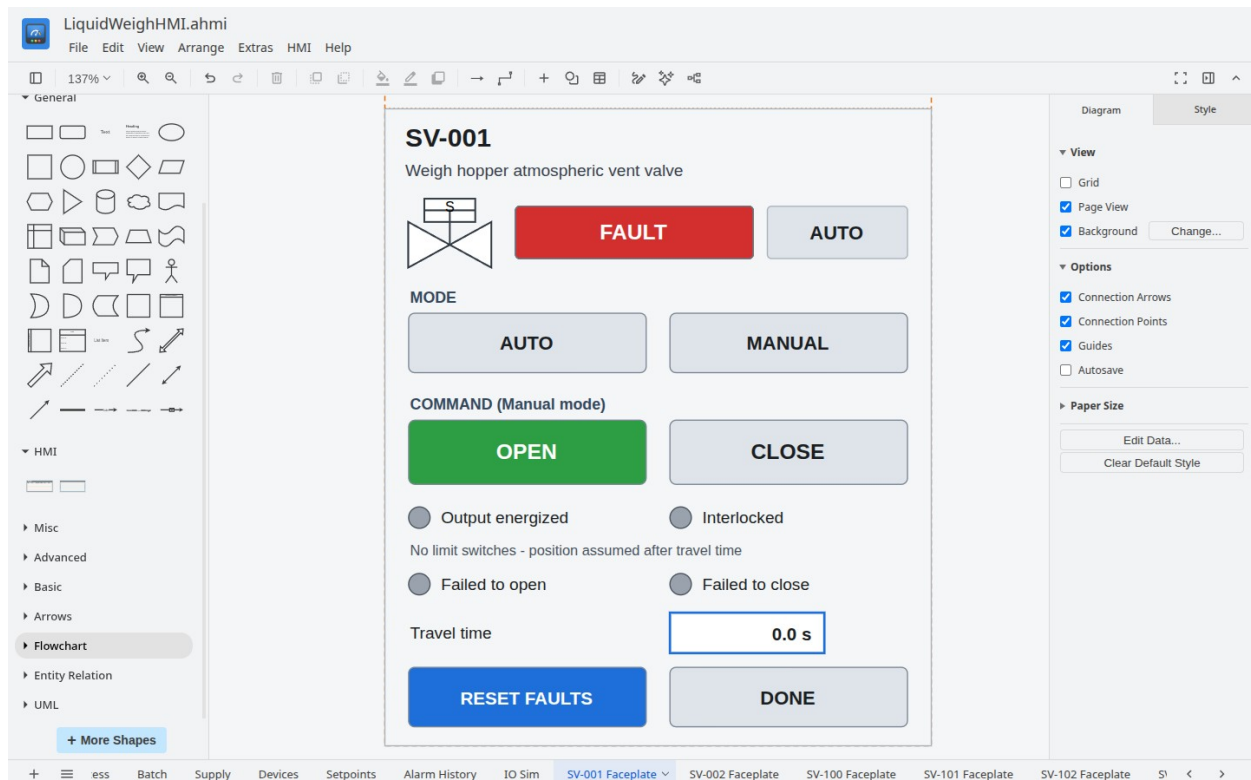


Figure 10. The SV-001 faceplate page in the editor

## 6.2 Navigation

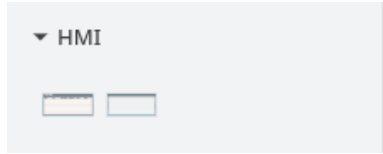
The navigation bar at the bottom of every LiquidWeighHMI screen is a row of buttons, each with a **Show Window** link to one screen. Because the main screens are Replace windows, opening one closes the current screen. Faceplates open the same way from the valve and pump symbols, and close with a **Hide Window** link on their DONE button.

## 7 Drawing objects

Screens are drawn with the editor's shapes: drag a shape from the Shapes panel onto the canvas, then set its fill, line and text in the Format panel. LiquidWeighHMI uses:

- Rectangles and rounded rectangles for panels, buttons and value boxes.
- Process symbols from **More Shapes > P&ID** (valves, pumps, vessels) and plain shapes for tanks and totes.
- Lines and connectors for piping.
- Text for labels and titles.

The **HMI** palette adds two objects: the **Alarm List**, the live list of active alarms at the bottom of every LiquidWeighHMI screen, and the **Alarm History**, used on the Alarm History screen.



*Figure 11. The HMI palette: Alarm List and Alarm History*

**Note:** Keep objects inside the Screen frame, and group repeated symbols (a valve with its tag label) so they can be copied with their links.

## 8 Animation links

Animation links make objects live. Select an object and open the **Animation** tab of the Format panel. The top of the tab lists every link type; click one to add it. Links already on the object are marked, and their settings appear below.

### Add animation link

#### TOUCH LINKS

User Input

Pushbutton ×

Action Script

Show Window

Hide Window

Horizontal

Vertical

#### LINE COLOR

Discrete

Analog

Discrete Alarm

Analog Alarm

#### FILL COLOR

Discrete ×

Analog

Discrete Alarm

Analog Alarm

#### TEXT COLOR

Discrete ×

Analog

Discrete Alarm

Analog Alarm

#### VALUE / MOVEMENT

Horizontal

Vertical

Width

Height

Horizontal

Vertical

Orientation

#### MISCELLANEOUS

Visibility

Blink

Enable ×

#### VALUE DISPLAY

Value Display

► Pushbutton

► Fill Color / Discrete

► Text Color / Discrete

Figure 12. The Animation tab for the START button, with its links marked

Table 8. Animation link families

| Family                           | Links                                                                                           | What they do                                                                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Touch links</b>               | User Input, Pushbutton, Action Script, Show Window, Hide Window, Slider Horizontal and Vertical | Operator actions: enter a value, set or toggle a tag, run a script, open or close a window, drag a slider |
| <b>Line, Fill and Text Color</b> | Discrete, Analog, Discrete Alarm, Analog Alarm                                                  | Color from an on/off value, from value bands, or from a tag's alarm state                                 |
| <b>Value / Movement</b>          | Location, Width, Height, Percent Fill, Orientation                                              | Move, size, fill or rotate an object in proportion to a value                                             |

| Family        | Links                     | What they do                                        |
|---------------|---------------------------|-----------------------------------------------------|
| Miscellaneous | Visibility, Blink, Enable | Show or hide, flash, and allow or block touch       |
| Value Display | Value Display             | Show a value or text in place of the object's label |

Numbers in links (ranges, limits) are expressions too, so they can come from tags. The sections below walk through the links LiquidWeighHMI uses.

## 8.1 A valve: status color, fault blink and faceplate

Each valve symbol on the Process screen carries three links. SV-102 is selected in Figure 13.

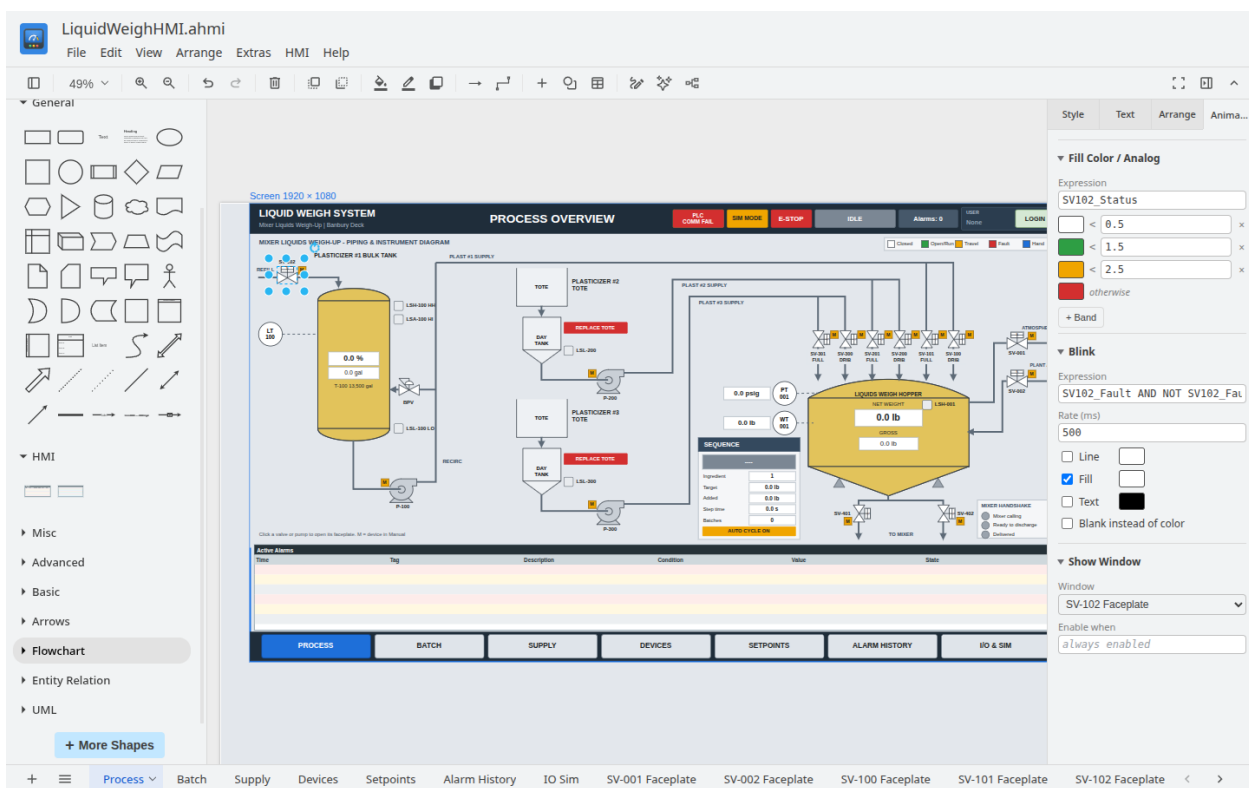


Figure 13. The SV-102 valve and its links

- **Fill Color / Analog** on SV102\_Status: below 0.5 (closed) white, below 1.5 (open) green, below 2.5 (travel) amber, otherwise (fault) red. The first band whose limit the value is below wins.
- **Blink** on SV102\_Fault AND NOT SV102\_FaultAck: flashes the fill while a fault is unacknowledged.
- **Show Window** SV-102 Faceplate: touching the valve opens its faceplate.

## 8.2 The weigh hopper: percent fill

The hopper vessel fills with its net weight. **Percent Fill / Vertical** maps WT001 from 0 to 150 lb to 0 to 100 % fill, from the bottom up.

The image shows a software configuration window with four tabs at the top: 'Style', 'Text', 'Arrange', and 'Anima...'. The 'Style' tab is selected. Below the tabs is a section titled '▼ Percent Fill / Vertical'. This section contains six input fields arranged vertically:

- 'Expression' with the value 'WT001'.
- 'Value at minimum' with the value '0'.
- 'Value at maximum' with the value '150'.
- 'Percent at minimum' with the value '0'.
- 'Percent at maximum' with the value '100'.

*Figure 14. Percent Fill on the weigh hopper*

**Note:** Percent Fill draws only the filled part of the object, outline included. Place an identical shape with no fill on top as the vessel outline, as LiquidWeighHMI does.

### 8.3 Hopper pressure: value display and alarm colors

The pressure box shows PT001 with one decimal and the suffix psig (**Value Display**), and its fill follows the tag's alarm state (**Fill Color / Analog Alarm**): normal white, High amber, HiHi red. The limits come from the tag in the Tag Dictionary, so they are set in one place.

StyleTextArrangeAnima...

▼ Value Display

Kind

Analog▼

Expression

PT001

Format

0.0

Prefix

text before value

Suffix

psig

▼ Fill Color / Analog Alarm

Tag

PT001

LoLo

Low

Normal

High

HiHi

Limits come from the tag in the dictionary

Figure 15. Value Display and Fill Color / Analog Alarm on PT001

## 8.4 Setpoints: user input

On the Batch screen the operator enters the batch volume by touching its value. **User Input** (analog) on Rcp\_TotalVol accepts 0 to 12 gallons with the prompt "Total batch volume (gal, max 12)" and shows an on-screen keypad. A **Value Display** on the same object shows the current value.

StyleTextArrangeAnima...

▼ Value Display

Kind

Analog

Expression

Rcp\_TotalVol

Format

0.00

Prefix

text before value

Suffix

gal

▼ User Input

Kind

Analog

Tag

Rcp\_TotalVol

Minimum

0

Maximum

12

Prompt

Total batch volume (gal, max 12)

☒ On-screen keypad

Figure 16. User Input and Value Display on the batch volume

## 8.5 Commands: pushbutton and enable

The START button on the Batch screen is a **Pushbutton** that sets PB\_Start when touched, allowed only while Seq\_Idle AND Seq\_StartPerm (Enable when). Its fill and text colors show when it is available. An **Enable** link on \_AccessLevel > 0 blocks it until someone is logged in (chapter 10).

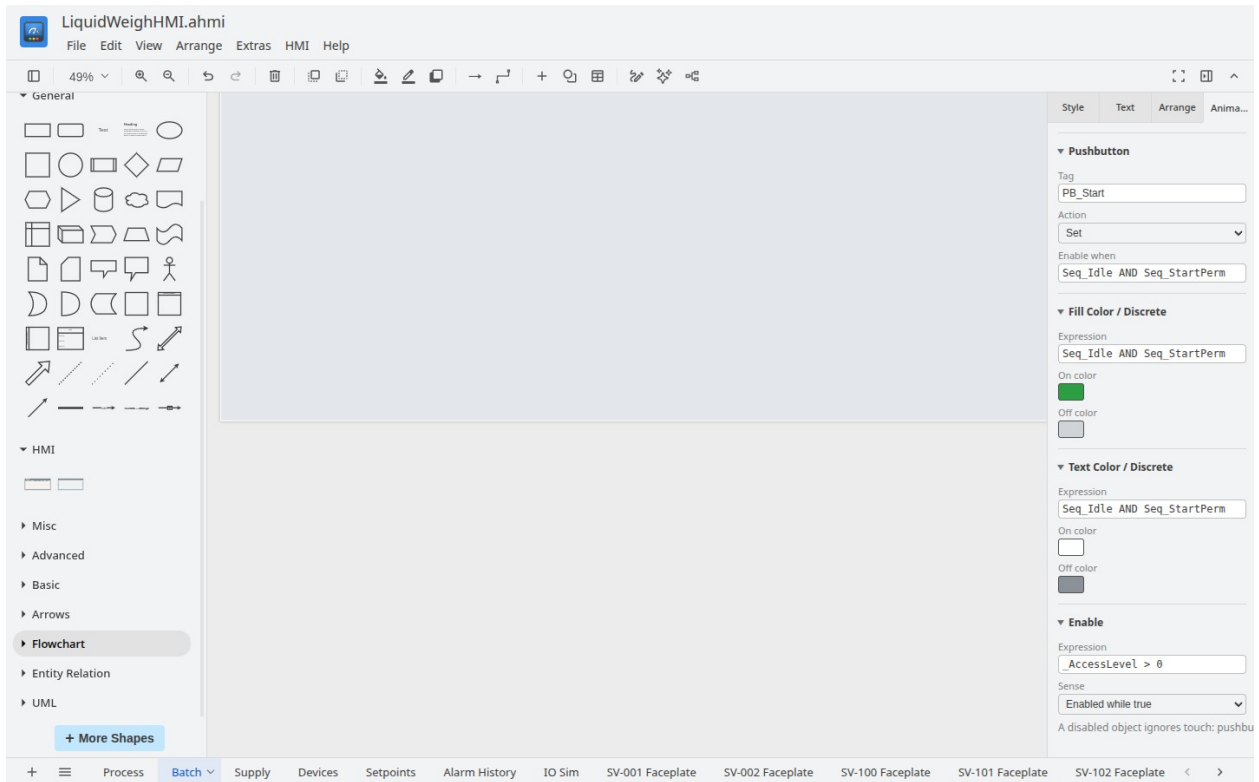


Figure 17. The START button: Pushbutton, colors and Enable

Table 9. Pushbutton actions

| Action        | Effect                                           |
|---------------|--------------------------------------------------|
| <b>Toggle</b> | Inverts the tag on each touch                    |
| <b>Set</b>    | Writes 1 (START, HOLD, RESUME in LiquidWeighHMI) |
| <b>Reset</b>  | Writes 0                                         |
| <b>Direct</b> | 1 while pressed, 0 when released (jog buttons)   |

## 9 Expressions and scripts

Links use expressions, and scripts (Action Script links and window scripts) use statements, both in the InTouch QuickScript style.

Table 10. Expression syntax

| Element                   | Examples                                                                    |
|---------------------------|-----------------------------------------------------------------------------|
| <b>Tags and dotfields</b> | PT001, SV001_Status, PT001.InAlarm, PT001.Acked, PT001.MaxEU                |
| <b>Operators</b>          | + - * / MOD, == <> < <= > >=, AND OR NOT, parentheses                       |
| <b>Literals</b>           | 12.5, "IDLE"                                                                |
| <b>Functions</b>          | Abs, Sqrt, Int, Round, Min(a, b), Max(a, b), StringLen, Text(value, "0.00") |

| Element              | Examples                                       |
|----------------------|------------------------------------------------|
| Statements (scripts) | Pump_Run = 1; IF cond THEN ... ELSE ... ENDIF; |

Table 11. System tags

| Tag            | Value                                                                  |
|----------------|------------------------------------------------------------------------|
| _AlarmsActive  | Number of active alarms (the Alarms: n box in LiquidWeighHMI's header) |
| _AlarmsUnacked | Number of unacknowledged alarms                                        |
| _AckAll        | Write 1 to acknowledge every alarm                                     |
| _Username      | Logged-in user, or None                                                |
| _AccessLevel   | Logged-in user's access level, or 0                                    |

## 9.1 Window scripts

A window can run a script when it opens (**On show**), repeatedly while it is open (**While showing**, every n ms) and when it closes (**On hide**). Every LiquidWeighHMI screen turns the PLC's sequence step number into the text shown in the header, four times a second:

```
IF Seq_Step == 0 THEN StepText = "IDLE"; ENDIF;
IF Seq_Step == 5 THEN StepText = "TARE"; ENDIF;
IF Seq_Step == 11 THEN StepText = "PLAST #1 FAST FILL"; ENDIF;
...
```

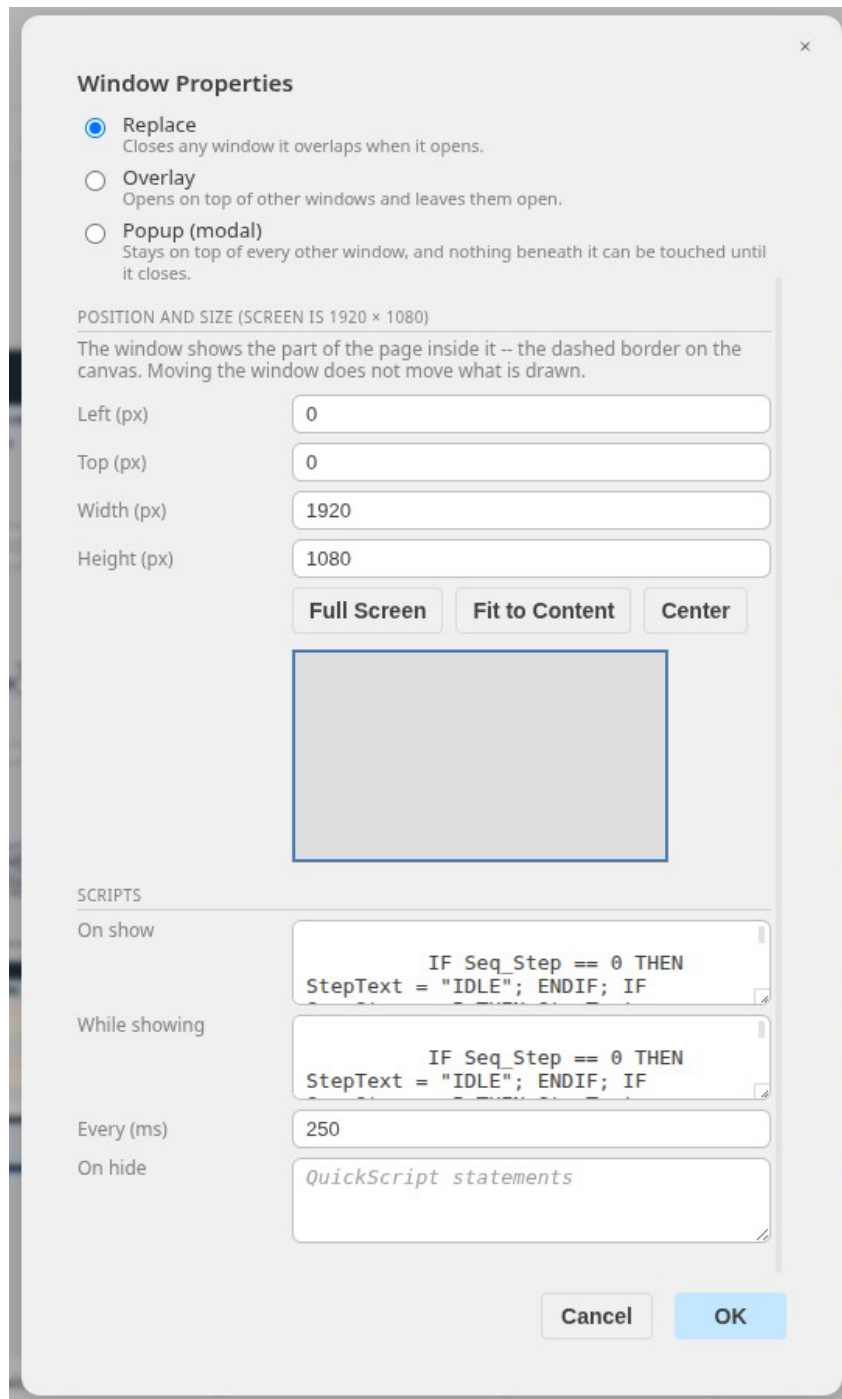


Figure 18. Window scripts of the Process screen (While showing every 250 ms)

## 9.2 Action scripts

An **Action Script** link runs statements when the object is pressed (On down), repeatedly while held (While down) and when released (On up). The LOGIN button in LiquidWeighHMI's header logs in or out:

```
IF _Username == "None" THEN ShowLogin(); ELSE Logout(); ENDIF;
```

| Style                                                                                | Text | Arrange | Anima... |
|--------------------------------------------------------------------------------------|------|---------|----------|
| <b>▼ Action Script</b>                                                               |      |         |          |
| On down<br><pre>IF _Username == "None" THEN ShowLogin(); ELSE Logout(); ENDIF;</pre> |      |         |          |
| While down<br><i>QuickScript statements</i>                                          |      |         |          |
| Every (ms)<br><input type="text" value="1000"/>                                      |      |         |          |
| On up<br><i>QuickScript statements</i>                                               |      |         |          |
| <b>▼ Value Display</b>                                                               |      |         |          |
| Kind<br><input type="text" value="Discrete"/>                                        |      |         |          |
| Expression<br><input none\""="" type="text" value="_Username &lt;&gt; \"/>           |      |         |          |
| On text<br><input type="text" value="LOGOUT"/>                                       |      |         |          |
| Off text<br><input type="text" value="LOGIN"/>                                       |      |         |          |
| Prefix<br><input type="text" value="text before value"/>                             |      |         |          |
| Suffix<br><input type="text" value="text after value"/>                              |      |         |          |
| <b>▼ Fill Color / Discrete</b>                                                       |      |         |          |
| Expression<br><input none\""="" type="text" value="_Username &lt;&gt; \"/>           |      |         |          |

Figure 19. The LOGIN button: Action Script, Value Display and colors

## 10 Alarms and security

### 10.1 Alarms

Alarms come from the limits set on tags (chapter 5). While the application runs, every alarmed tag is watched, whether or not a screen shows it:

- Entering alarm, or escalating (High to HiHi), raises an alarm that needs acknowledging.
- An alarm stays in the Alarm List until it has returned to normal **and** been acknowledged (ISA-18.2).

- The Alarm List object has an **Ack All** button and an **Ack** button on each row. A script can acknowledge with `_AckAll = 1;` or `Tag.Acked = 1;`.
- Every alarm event is written to a daily history file kept for 90 days; the Alarm History object shows it (Figure 30).

## 10.2 Users and access levels

Open **HMI > Users** to define who can log in. Each user has a name, a password (stored only as a salted hash) and an access level from 0 to 9999.

**Users**

Access levels run from 0 to 9999. Animations read the logged-in user through `_Username` and `_AccessLevel`; `ShowLogin()` in a script opens the login window.

| User        | Access Level | Edit | Remove |
|-------------|--------------|------|--------|
| Operator    | 1000         | Edit | Remove |
| Maintenance | 2000         | Edit | Remove |
| Engineer    | 9999         | Edit | Remove |

USER

Name:

Access level:

Password:

Confirm password:

New Add

Cancel OK

Figure 20. The Users dialog with LiquidWeighHMI's three users

Table 12. LiquidWeighHMI users

| User        | Access level | Can                                            |
|-------------|--------------|------------------------------------------------|
| Operator    | 1000         | Run batches                                    |
| Maintenance | 2000         | Also operate devices in manual                 |
| Engineer    | 9999         | Everything, including setpoints and simulation |

Screens use `_AccessLevel` to protect controls: an **Enable** link such as `_AccessLevel >= 2000` blocks touch on an object, and a **Visibility** link can hide it. The script function `ShowLogin()` opens the built-in login window (Figure 21); `Login(name, password)`, `Logout()`, `ChangePassword(old, new)` and `ShowUserManager()` support custom login screens.

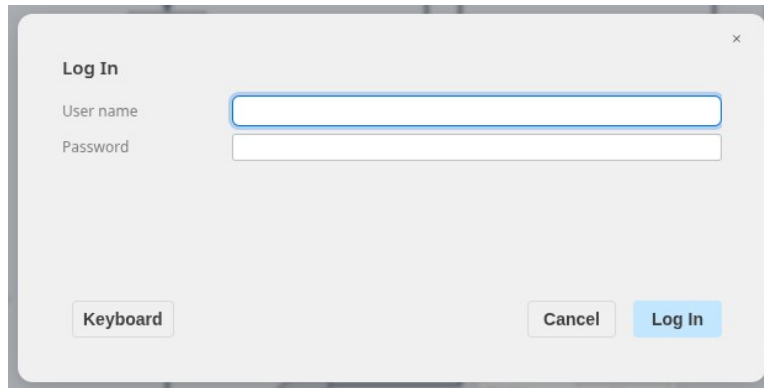


Figure 21. The login window opened by ShowLogin()

## 11 Testing

### 11.1 Validate

**HMI > Validate Expressions** checks every link, script and PLC address and lists any problem with the page and object; click a problem to go to it. Validate before every Run and Publish.



Figure 22. Validation of LiquidWeighHMI: no problems found

### 11.2 Run

**HMI > Run** (F5) runs the application inside the Studio: the drawing is locked, the start-up windows open, and the application connects to its PLCs. A green RUNNING banner shows that the screen is live and that touching it operates real equipment. **HMI > Stop** returns to editing.

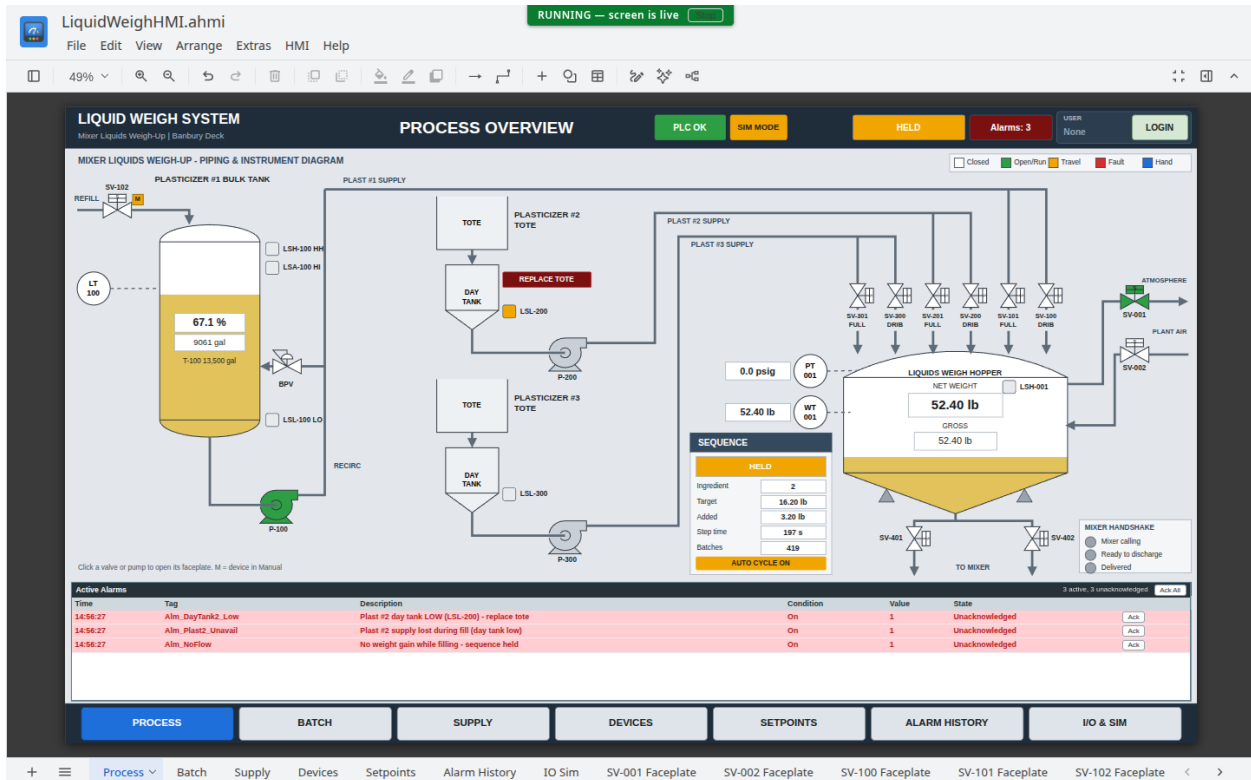


Figure 23. LiquidWeighHMI running in the Studio

The screens below were captured while LiquidWeighHMI ran against its controller, with a batch held on a plasticizer supply alarm.

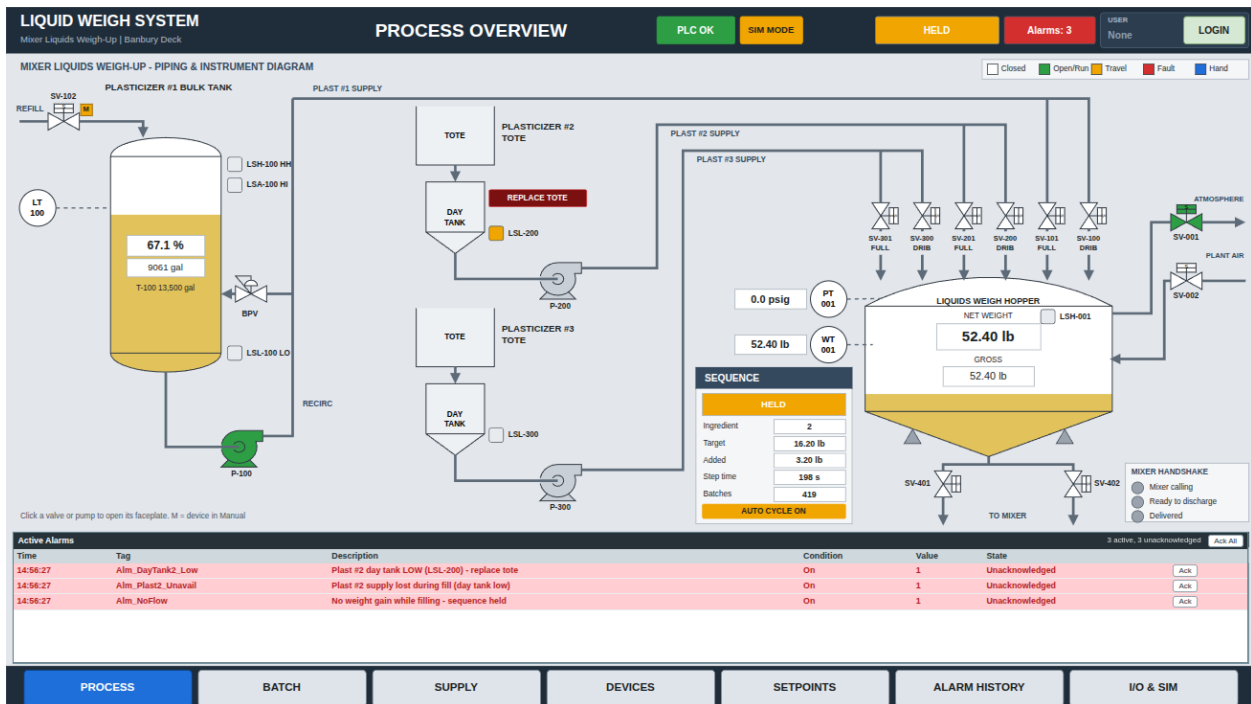


Figure 24. Process screen: tank and hopper levels, valve states, sequence and active alarms

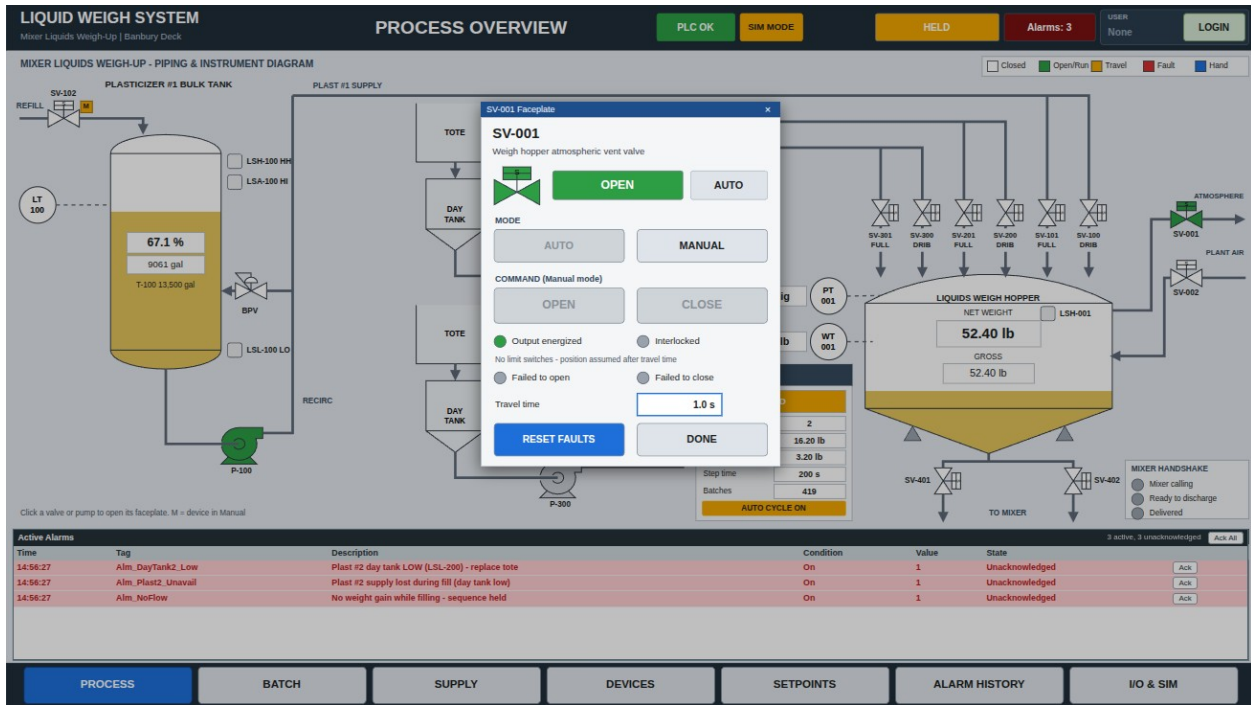


Figure 25. SV-001 faceplate: mode, commands, interlocks and fault reset

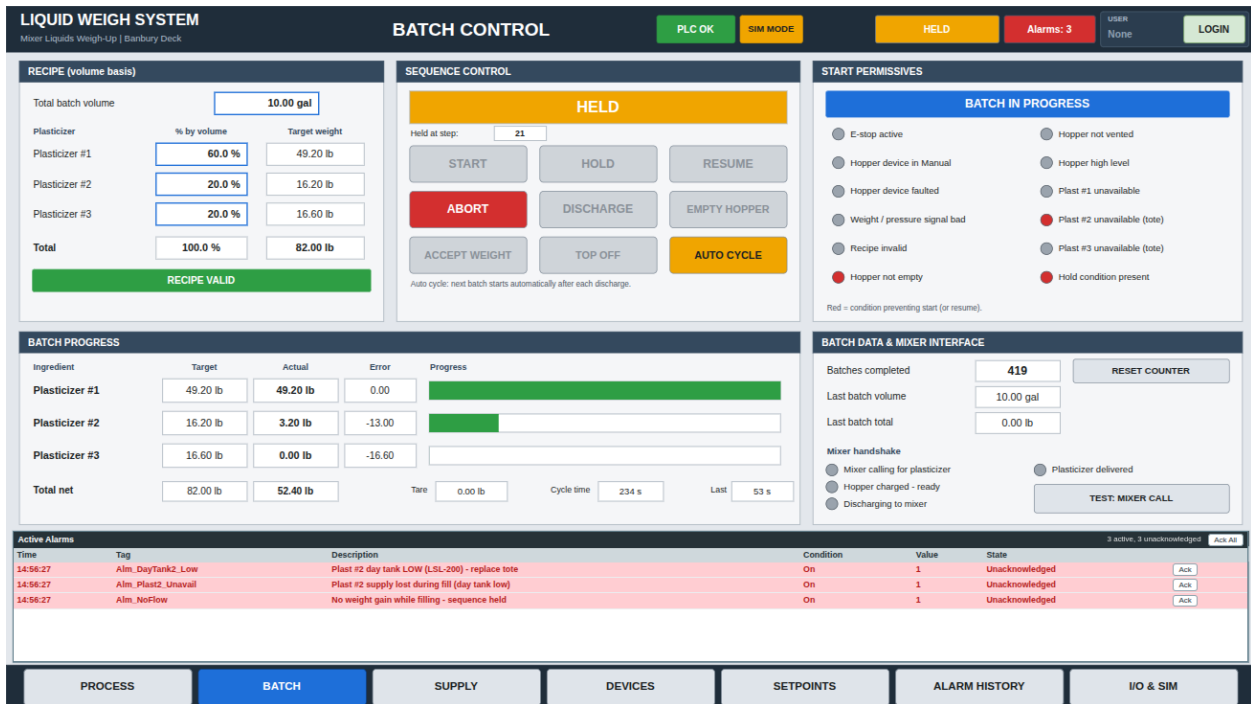


Figure 26. Batch screen: recipe, sequence control, start permissives and batch progress

LIQUID WEIGH SYSTEM

Mixer Liquids Weigh-Up | Banbury Deck

SETPOINTS

PLC OK

SIM MODE

HELD

Alarms: 3

user: None

LOGIN

MATERIALS

|                         | Plast #1 | Plast #2 | Plast #3 |
|-------------------------|----------|----------|----------|
| Density (lb/gal)        | 8.20     | 8.10     | 8.30     |
| Dribble weight (lb)     | 3.00     | 3.00     | 3.00     |
| Preact / in-flight (lb) | 0.19     | 0.03     | 0.02     |

Dribble weight: switch from full flow to dribble this far before target.  
Preact: close the dribble valve this far before target.

Automatic preact adaptation ON

Preact adaptation gain 0.50

Maximum preact 2.00 lb

Maximum batch volume 12.0 gal

Minimum ingredient target 0.10 lb

Recirc pump runs in Auto ON

LOAD DEFAULTS Reloads all setpoints and the recipe from PLC defaults.

WEIGHING

Ingredient tolerance (+/-) 0.50 lb

Scale settle time 2.0 s

No-flow alarm time 10.0 s

No-flow minimum gain 0.20 lb

Hopper empty weight 1.00 lb

Auto-cycle restart delay 2.0 s

Day tank low pump inhibit delay 300 s

Tolerance: ingredient within +/- this weight passes; otherwise the sequence waits for ACCEPT or TOP OFF on the Batch screen.

No-flow: fill is held if weight does not rise by the minimum gain within the alarm time.

PRESSURE & TIMING

Fill valves interlocked above 3.0 psig

Hopper vented below 2.0 psig

Charge pressure 40.0 psig

Charge timeout 30 s

Discharge complete below 10.0 psig

Discharge complete debounce 2.0 s

Discharge timeout 120 s

Vent close delay 1.0 s

Vent timeout 30 s

Discharge ends when the air blows through the empty hopper and pressure falls below 'Discharge complete' for the debounce time.

Active Alarms

| Time     | Tag                | Description                                     | Condition | Value | State          |                  |
|----------|--------------------|-------------------------------------------------|-----------|-------|----------------|------------------|
| 14:56:27 | Alm_DayTank2_Low   | Plast #2 day tank LOW (LSL-200) - replace tote  | On        | 1     | Unacknowledged | <span>ACK</span> |
| 14:56:27 | Alm_Plast2_Unavail | Plast #2 supply lost during fill (day tank low) | On        | 1     | Unacknowledged | <span>ACK</span> |
| 14:56:27 | Alm_NoFlow         | No weight gain while filling - sequence held    | On        | 1     | Unacknowledged | <span>ACK</span> |

PROCESS

BATCH

SUPPLY

DEVICES

SETPOINTS

ALARM HISTORY

I/O & SIM

Figure 27. Setpoints screen: materials, weighing, pressure and timing

LIQUID WEIGH SYSTEM

Mixer Liquids Weigh-Up | Banbury Deck

DEVICES

PLC OK

SIM MODE

HELD

Alarms: 3

user: None

LOGIN

VALVES & PUMPS

| Device | Description                           | Mode   | State   | Interlocks | Command | Fault |                           |        |           |
|--------|---------------------------------------|--------|---------|------------|---------|-------|---------------------------|--------|-----------|
| SV-001 | Weigh hopper atmospheric vent valve   | AUTO   | OPEN    | OK         | ON      |       | AUTO                      | MANUAL | FACEPLATE |
| SV-002 | Weigh hopper pressurizing (air) valve | AUTO   | CLOSED  | BLOCKED    | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-100 | Plast #1 dribble flow valve           | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-101 | Plast #1 full flow valve              | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-102 | Plast #1 bulk tank refill valve       | MANUAL | CLOSED  | OK         | OFF     |       | Operator (refill station) |        | FACEPLATE |
| SV-200 | Plast #2 dribble flow valve           | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-201 | Plast #2 full flow valve              | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-300 | Plast #3 dribble flow valve           | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-301 | Plast #3 full flow valve              | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-401 | Weigh hopper discharge valve #1       | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| SV-402 | Weigh hopper discharge valve #2       | AUTO   | CLOSED  | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| P-100  | Plast #1 recirculation pump           | AUTO   | RUNNING | OK         | ON      |       | AUTO                      | MANUAL | FACEPLATE |
| P-200  | Plast #2 tote pump                    | AUTO   | STOPPED | BLOCKED    | OFF     |       | AUTO                      | MANUAL | FACEPLATE |
| P-300  | Plast #3 tote pump                    | AUTO   | STOPPED | OK         | OFF     |       | AUTO                      | MANUAL | FACEPLATE |

ALL HOPPER DEVICES AUTO
ALL HOPPER DEVICES MANUAL
RESET FAULTS

Manual mode lets the operator open/close or start/stop a device from its faceplate. Interlocks still apply. Putting a hopper device in Manual holds a running batch.

Active Alarms

| Time     | Tag                | Description                                     | Condition | Value | State          |                  |
|----------|--------------------|-------------------------------------------------|-----------|-------|----------------|------------------|
| 14:56:27 | Alm_DayTank2_Low   | Plast #2 day tank LOW (LSL-200) - replace tote  | On        | 1     | Unacknowledged | <span>ACK</span> |
| 14:56:27 | Alm_Plast2_Unavail | Plast #2 supply lost during fill (day tank low) | On        | 1     | Unacknowledged | <span>ACK</span> |
| 14:56:27 | Alm_NoFlow         | No weight gain while filling - sequence held    | On        | 1     | Unacknowledged | <span>ACK</span> |

PROCESS

BATCH

SUPPLY

DEVICES

SETPOINTS

ALARM HISTORY

I/O & SIM

Figure 28. Devices screen: every valve and pump with mode, state and faceplate

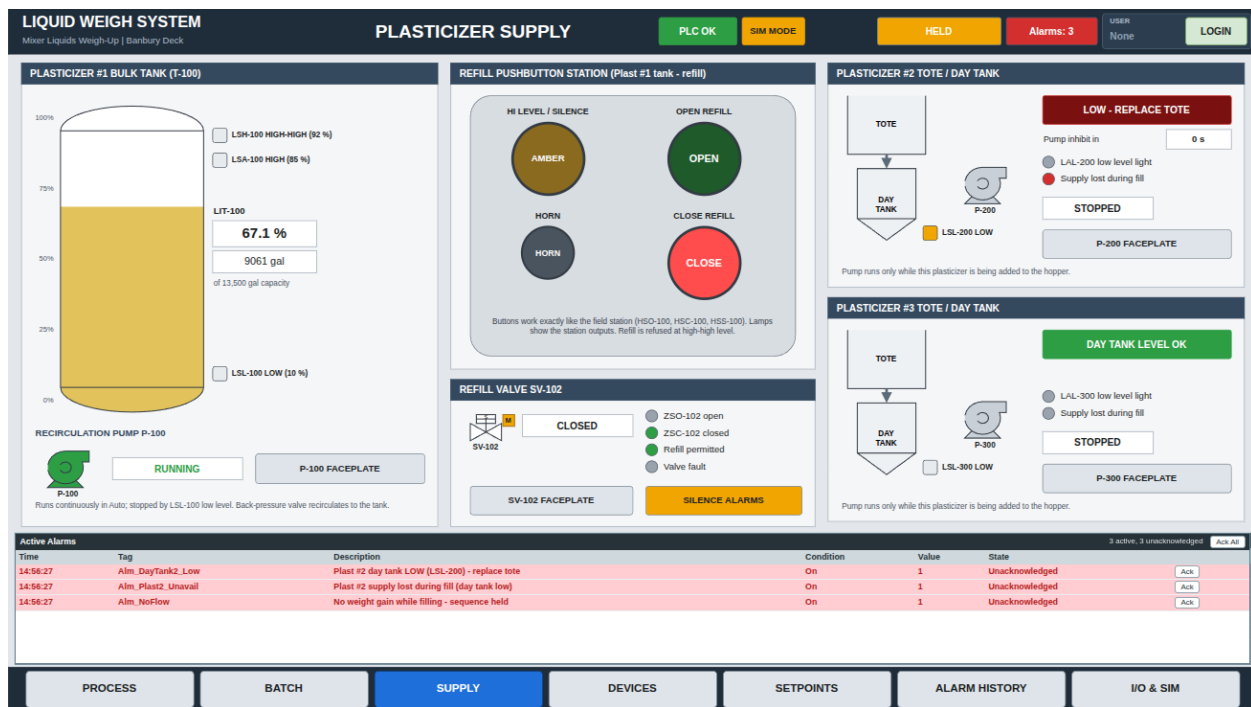


Figure 29. Supply screen: plasticizer tank, refill station and totes

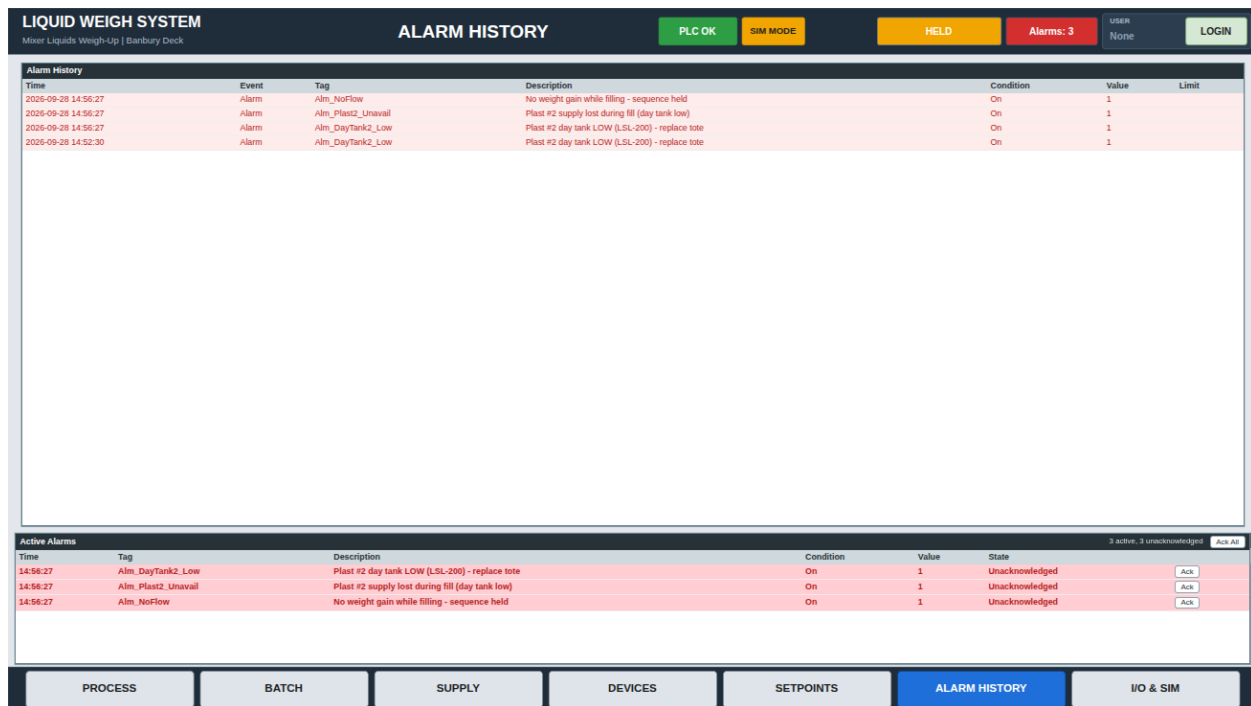


Figure 30. Alarm History screen

**Note:** HMI > Runtime Log lists connections, device errors, writes and script errors from the run.

## 12 Deploying

An application can reach the plant floor in three ways.

*Table 13. Deployment options*

| Option                    | What the target gets                                                                               | Best for                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>HMI &gt; Publish</b>   | A Windows installer of this one application, with its own name and icon, starting straight into it | A dedicated, locked-down operator station                        |
| <b>Append HMI Desktop</b> | One runner installed once; runs any .ahmi file, from shortcuts or at log-in                        | Stations whose projects change often; Linux                      |
| <b>Append HMI Web</b>     | A web server; operators open the application in any browser                                        | Viewing and operating from tablets, laptops and control-room PCs |

### 12.1 Publish

**HMI > Publish** builds a Windows installer from the open project. Give the product name, version and publisher, choose an icon, whether it installs for all users, and whether it adds a desktop shortcut or starts with Windows. Run the Setup.exe on the target PC; the application then starts full screen as set in Application Settings.

**Publish**

Creates a Windows installer that runs this project full time on the target PC, with the comms server. Window mode and exit are set in Application Settings.

PACKAGE

Product name: LiquidWeighHMI

Version: 1.0.3

Publisher: Optional

Icon: Default Browse...

INSTALLER

Install for: All users (needs administrator) ▼

Desktop shortcut: ☒

Start with Windows: ☐

Compression: Smaller installer (a few minutes) ▼

Save to: C:\Users\engineer\Documents Browse...

Cancel Publish

Figure 31. The Publish dialog

## 12.2 Append HMI Desktop

Append HMI Desktop runs .ahmi files without the editor. Choose an application, then **Run now**, create a desktop shortcut or menu entry, or tick **Run this application automatically when I log in**. From the command line: `append-hmi-desktop LiquidWeighHMI.ahmi`.

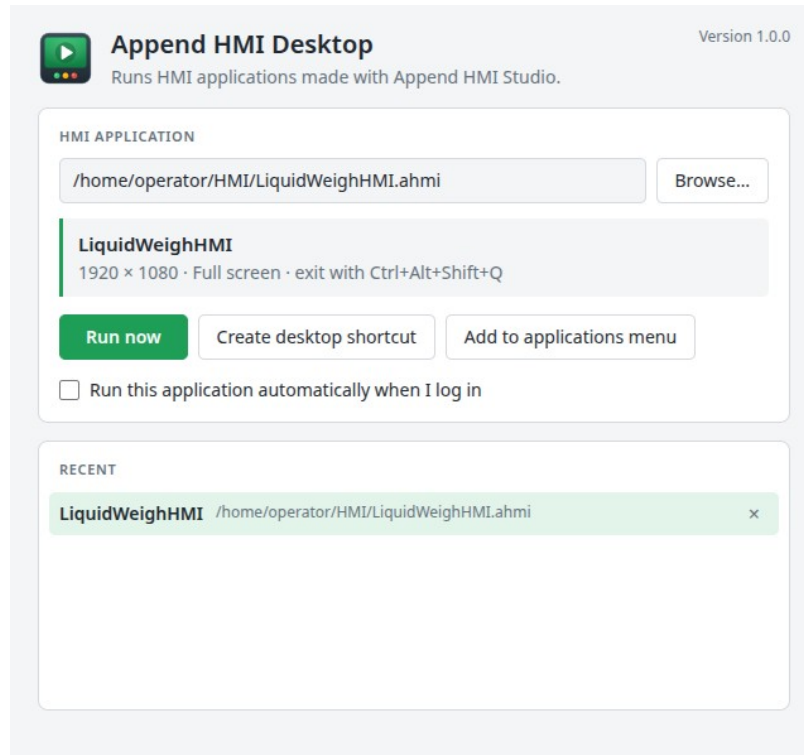


Figure 32. The Append HMI Desktop launcher with LiquidWeighHMI selected

### 12.3 Append HMI Web

Append HMI Web serves an application to web browsers on the plant network. Choose the application and a port (8480 by default), then **Start web server**; the launcher shows the link to share, with a **Copy** button. If the port is already in use on the PC (by another web server, or another application being served), choose another. In the browser, the view button in the corner fits the screen to the window, maximizes it, or shows it at its original size with scroll bars.

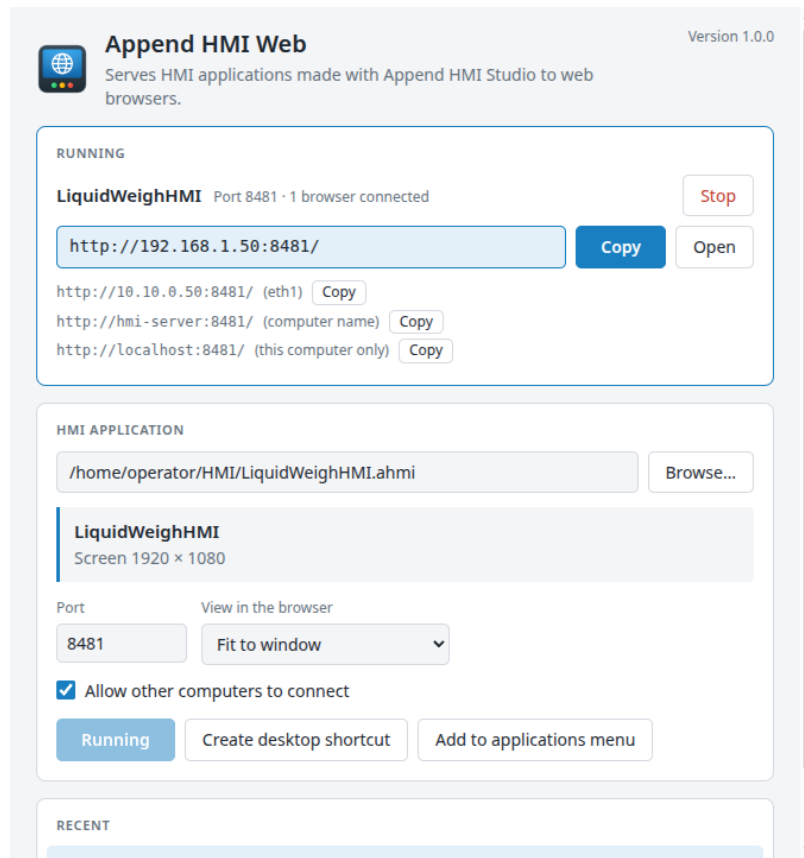


Figure 33. The Append HMI Web launcher serving LiquidWeighHMI on port 8481 to one browser

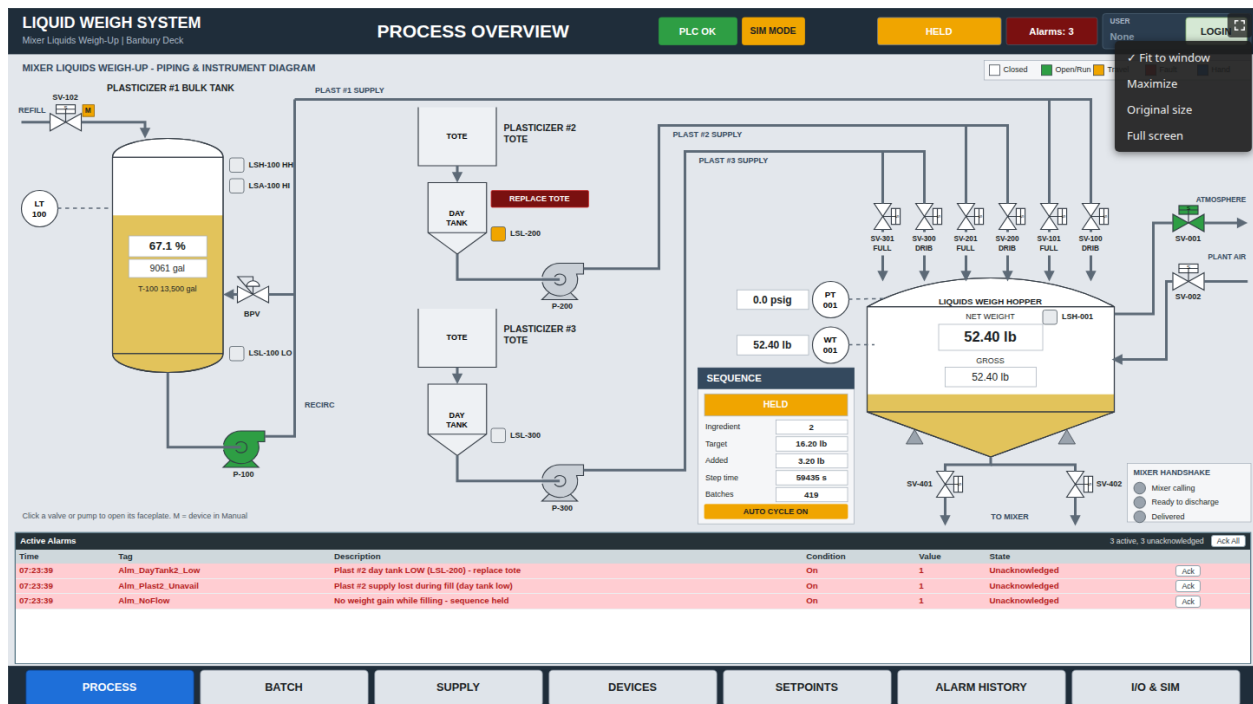


Figure 34. LiquidWeighHMI in a web browser, with the view menu open

## 13 Tips and troubleshooting

Table 14. Common problems

| Symptom                                  | Cause and remedy                                                                                                                        |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Values show ####                         | The tag has bad quality: the PLC is not connected or the address is wrong. Check HMI > Runtime Log and Test Connection in HMI > Devices |
| PLC COMM FAIL in the header              | The device is unreachable. Check the IP address, slot and network                                                                       |
| A button does nothing                    | An Enable or Pushbutton "Enable when" condition is false, or nobody with the access level is logged in                                  |
| Validate reports an expression error     | Expressions take no ";" (only script statements do); tag names must exist in the Tag Dictionary                                         |
| A popup opens in the wrong place         | Its objects must be drawn at the window's position on its page (Window Properties > Left, Top)                                          |
| Retentive values are not what you expect | Use HMI > Clear Retentive Values to start from the initial values again                                                                 |

Build a symbol once, with its links, group it, and copy it: links travel with the copy. The Animation tab can also copy links from one object and paste them onto others.

## Appendix A Command-line automation

Append HMI Studio can build, check, render and publish applications without its window, for scripting and for AI agents:

Table 15. Command-line modes

| Command                                                                            | Does                                               |
|------------------------------------------------------------------------------------|----------------------------------------------------|
| <code>--hmi-build spec.json -o app.ahmi</code>                                     | Builds a project from a JSON description           |
| <code>--hmi-check app.ahmi</code>                                                  | Validates it (exit code 2 when problems are found) |
| <code>--hmi-render app.ahmi -o folder</code>                                       | Renders every page to PNG                          |
| <code>--hmi-dump app.ahmi -o spec.json</code>                                      | Writes a project back out as JSON                  |
| <code>--hmi-publish app.ahmi --product "Name" --app-version 1.0.0 -o folder</code> | Builds the Windows installer                       |

The JSON format is documented in doc/HMI\_AUTOMATION.md in the Append HMI Studio repository.